

HENRY CHU

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Education

University of Toronto | BAsC - Mechanical Engineering (Co-op) | Toronto, ON Sep 2021 - Present

- Awards: Skoll Scholarship, Bennett Scholar, Engineering Society Award, Dean's Honours List
- Specialization: Mechatronics and Energy & Environment stream, Minor in Mechatronics & Robotics

Work Experience

Mechanical Engineering Designer | Safety Power | Mississauga, ON May 2024 - Aug 2025

- Designed Safety Power's SCR after-treatment systems for diesel generators for 30+ projects across Canada and the United States using DfMA principles with SolidWorks and NanoCAD, supervised by P.Eng
- Completed a custom equipment design for a retrofit project and oversaw fabrication and installation from start to finish
- Presented proposals and design solutions to senior management for approval regarding the retrofit project
- Performed on-site equipment inspections, testing and repairs at fabrication shops in Georgetown and Waterloo, Ontario
- Prepared as-builts, BOMs and shop drawings for RFQs and RFPs to subcontractors and clients
- Developed a Python-based shop drawing pdf search and reader tool to streamline team access to past project designs
- Worked in close liaison with external contractors and led internal coordination meetings to track project schedule

Structural Defect Analyst | RaSpect Intelligence Inspection Limited | Hong Kong SAR Jul - Sep 2023

- Analyzed and reported structural defects (e.g. crack, delamination, honeycombing) of residential building facade and highway infrastructure in Hong Kong using RaSpect's drone building inspection and AI software

Civic Engine Plant Manufacturing Intern | Honda Canada Manufacturing | Alliston, ON May - Jul 2023

- Conducted quality inspections for casting defects (e.g. galling, flash, blowholes, shrinkage) in the Civic IC engine
- Safely operated the high-pressure and low-pressure die-casting machines to produce engine blocks and heads

Project Experience

Biometric Vehicle Immobilizer System Design and Prototype (NDA) - Capstone Client Contact Sep - Apr 2026

- Collaborating with an industrial client to design and test new system enclosures for electronic components to ensure compliance with automotive and safety standards
- Liaising with an industrial client and supervisor to propose design changes, collect feedback, and schedule team meetings

Robotics Control Programming and Data Analysis for Autonomous Mobile Manipulator Robot Jan - Apr 2026

- Programming an autonomous home robot (Turtlebot 4) mounted with a 5 d.o.f. manipulator using ROS2 and C++ to complete spatial exploration and object manipulation tasks
- Analyzing sensor data and troubleshooting LiDAR and hazard detection mechanisms of Turtlebot

CFD Simulation of Airflow Around a Spoiler - SimScale Project April 2025

- Performed an incompressible fluid flow simulation around a spoiler using SimScale and learnt fundamentals of CFD

[Bored.com](#) - Harvard University CS50x Final Project

Oct - Dec 2024

- Developed a social networking web app aimed to connect users with shared interests and hobbies to organize/attend in-person events using Python, HTML, CSS, Google Maps API, and SQL

TradingSim - Harvard University CS50x Project

May - Oct 2024

- Developed a web app that simulates real-time stock portfolio management with updated stock prices using Python, HTML, CSS, IEX API, and SQL, and analyzed stock data to identify statistical trends

PCB Design and Fabrication for an Adjustable DC Power Supply

Jan - Apr 2024

- Designed and built a low-cost DC supply from scratch using Eagle and LTSpice that can power common circuits such as microcontrollers, motors, sensors, etc.

Sustainability and Life Cycle Analysis of Materials for Anti-Fatigue Mats: PU vs PE Foam

Jan - Apr 2024

- Conducted Streamlined Life Cycle Assessment using OpenLCA and inventory flow diagrams to identify key life stages and environmental stressors in polyurethane and polyethylene materials
- Conducted FOC, Economic, and Societal analyses and compared both materials in terms of overall environmental impacts across the entire life cycle

ANSYS FEA of Gears, Stringer, Dovetail, and Pressure Vessel - MIE222 Project

Jan - Apr 2023

- Conducted Finite Element Analyses (FEA) on aircraft components using ANSYS Workbench to analyze stress distribution and modes of deformation under stress, and experiment with different mesh settings
- Employed GD&T principles to create sketches and surface bodies on ANSYS before performing FEA

Leadership Experience

Mech and TrackOne Student Ambassador | U of T Engineering Recruitment Centre

Sep 2023 - Nov 2025

- Represented and promoted the faculty's programs at university fairs and outreach programs such as the OUF and Fall Campus Day
- Connected with 100+ high school students and parents in on-stage presentations and one-on-one conversations

Drone Hardware Engineer | U of T Aerospace Team - Unmanned Aerial Systems

Sep 2023 - Aug 2025

- Designed and fabricated drone payloads for the Aerial Evolution Association of Canada (AEAC) national competition
- Coordinated with multi-disciplinary subteams to track assembly progress and ensure competition timeline is met

Volunteering Experience

2026 Chinese New Year Gala Dinner Party Coordinator | Hong Kong-Canada Business Association

Feb 2026

- Coordinated with volunteers and project managers to setup the silent auction booths, dining hall and main stage
- Connected with high profile guests and event VIPs and sold lucky draw tickets

UofT Soccer Intramural Open Division 3 Passion FC Team Captain

Jan - Apr 2024

- Organized and led a non-faculty team into the final round of Division 3 Intramural Playoffs
- Team top goal scorer

Skills

English, Cantonese, Mandarin, Japanese, SolidWorks, NanoCAD, ANSYS, Python, C++, ROS2, MS Office Suite, Google Suite